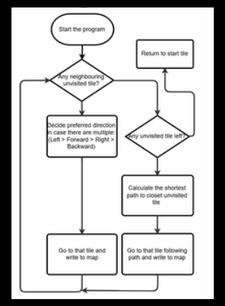
PCMS_ET

SOFTWARE

OVERVIEW

our programming is based on the ***Robot Operating System (ROS)***, an environment that provides files and services for communication between programs. This allows us to divide the tasks between several programs at the same time.

FLOW-CHART OF THE MAIN LOOP



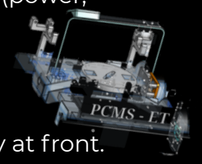
CIRCUIT DESIGN

SUB BOARD

- Controls ***10*** laser distance sensors and a gyroscope via I2C
- Aggregates sensor readings and sends data to the main board through ***UART***

MAIN BOARD

Central controller with reserved ***PCB interfaces***:
- ***2*** motor ports (2.54P), driving ***4*** motors (2 front, 2 rear)
- ***3*** I2C ports, ***4*** ADC inputs, and ***4*** GPIO pins
- ***2*** UART connections (sub board + camera module)
- Powered by a ***12V*** battery with a ***12V–5V*** converter for board operation
- Includes an ***OLED*** display and three buttons (power, motor enable, start)



HARDWARE

SENSOR & ACTUATOR INTEGRATION

- Path tracking: ***9 Laser Sensor VL53L0X*** array at front.
- Victim recognition: ***Vision modules*** (Sipeed MaixCAM) on both sides.
- Terrain perception: ***Color sensor*** (bottom) for forbidden zones; ***bumper switches*** (front sides) for collision prevention.
- Drive system: ***25GA-370 DC geared motors*** with encoders for odometry.

CONTROL & CIRCUIT SYSTEM

- ***ESP32*** main controller.
- ***Two custom PCBs*** for power management, signal transmission, and sensor integration, reducing wiring issues and improving ***anti-interference***.

CHASSIS & SUSPENSION

- Ground clearance: ***25 mm***.
- ***Rocker-bogie (parallel linkage) suspension*** for terrain adaptability.
- Battery placed at ***center-bottom*** for ***low CG*** and ***balanced weight distribution***.

RESCUE KIT DEPLOYMENT SYSTEM

- ***Dual-slide track*** with a ***central rotating disk*** for left/right delivery.
- ***Bottom guide rail*** ensures precise drop into victim zones.

WHEELS & TIRE DESIGN

- ***3D-printed resin hubs*** with ***tooth grooves*** for obstacle climbing
- ***Ring grooves*** lock ***silicone tires*** firmly, preventing slippage or detachment during turns/climbs.

LAO CHI IENG

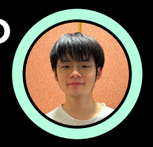
- Hardware
- Traversing alorithm

LUI IEK TONG

- Hardware
- Documentation

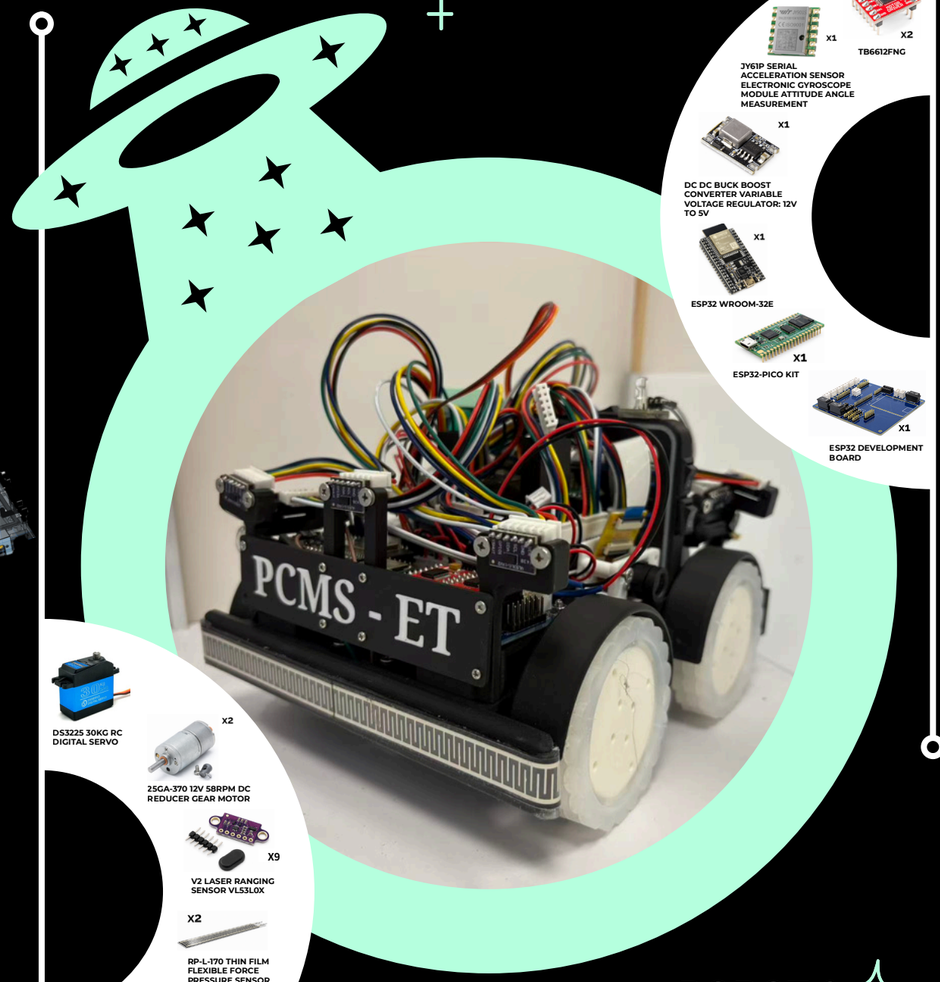
WONG CHI KIO

- Software
- Image recognition



LEI UN HOU

- Logs
- Sensor system



ROBOT & COMPONENT

EXPLORING ALL OF THE MAZE BY SHORTEST ROUTE

Base Method: We designed a maze exploration method that maintains a two-dimensional map of discovered tiles and selects robot actions based on its orientation and local information.
Baseline Reference and Its Limitation: As a reference, we applied a right-hand-based exploration strategy. When the number of maze tiles is N, this approach requires approximately *2.2N search actions on average*, resulting in inefficient exploration.
Proposed Solution: To improve efficiency, we developed an original algorithm that combines *depth-first search* and *breadth-first search*, named ***DBFS (Depth and Breadth First Search).***
Algorithm Mechanics: DBFS prioritizes movement toward the nearest reachable tile among known candidates. Robot behavior is divided into search and move, defined below.

SEARCH

Adjacent unexplored tiles are added to a ***LIFO*** structure named ***UNSEARCH*** with the priority:
[LEFT → FRONT → RIGHT]

MOVE

The robot selects the next destination from ***UNSEARCH***. Using ***BFS***, the robot computes the shortest path and runs a brief simulation before moving.

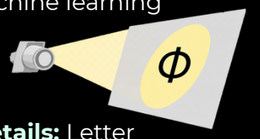
SEARCH AND MOVE ARE EXECUTED ALTERNATELY.
EXPLORATION ENDS WHEN UNSEARCH BECOMES EMPTY

RESULT

DBFS was evaluated against a right-hand-based method. The number of search actions was reduced by approximately ***33%***, improving exploration efficiency and enabling more reliable exit reach and ***Exit Bonus*** achievement

IMAGE RECOGNITION USING MACHINE LEARNING

Victims on maze walls are detected while the robot moves forward using ***UART communication*** with camera modules. Color victims are identified using ***LAB thresholding***, while letter victims are recognized by a trained machine learning model. White LEDs are used to reduce the influence of ambient lighting, improving recognition accuracy.



Machine Learning Details: Letter recognition uses a ***MobileNet-V2*** model trained on approximately ***6,000 self-collected images***. The model is converted to ***TensorFlow Lite*** and deployed for real-time inference

Training progress is monitored using learning curves to prevent overfitting, and recognition accuracy improves with continued training.